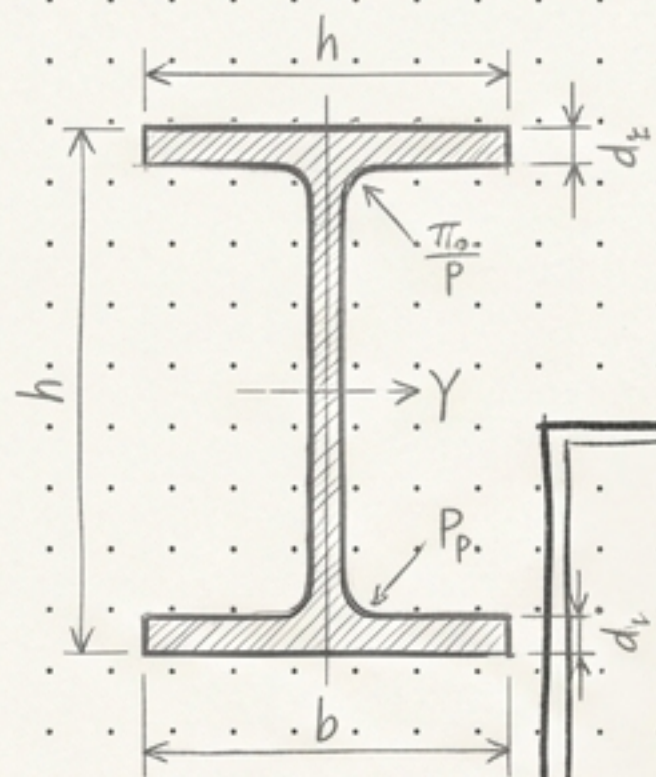
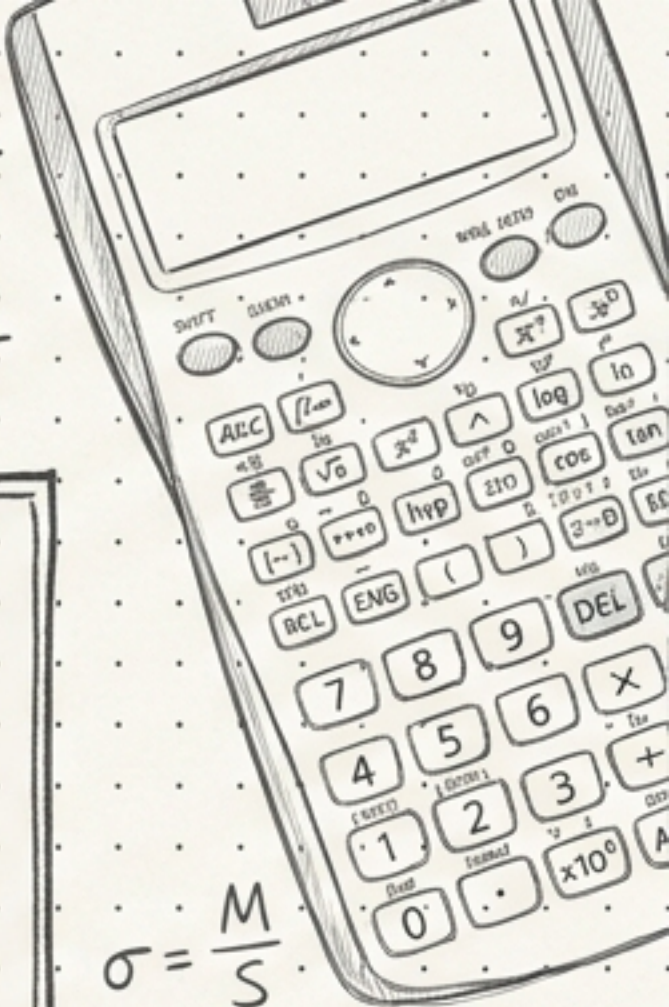




$$M = \frac{wL^2}{8}$$

$$\sigma = \frac{M}{S}$$

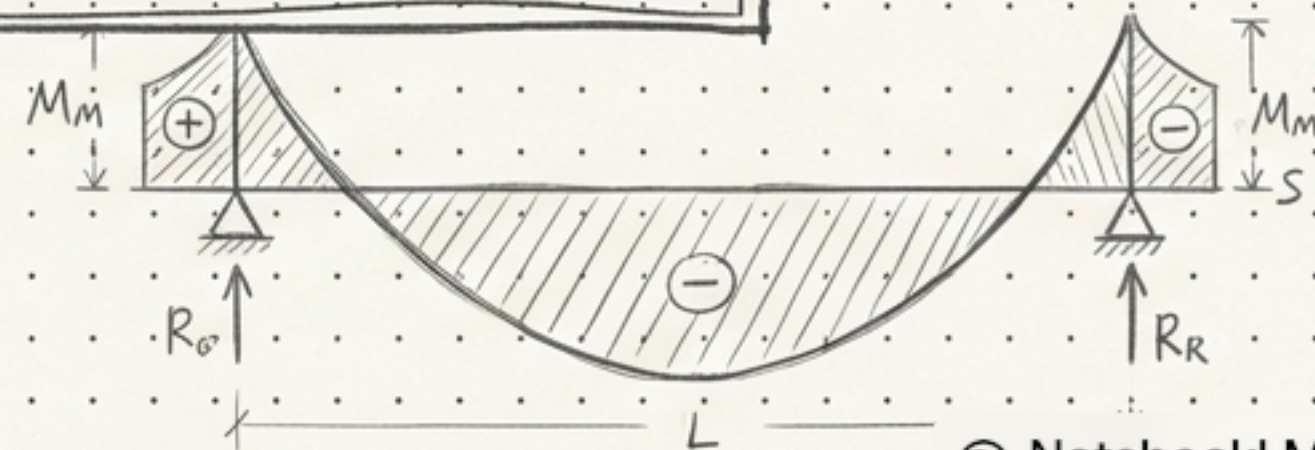


$$\sigma = \frac{M}{S}$$

SS-U1-2 梁桿件 | 深度複習

Steel Design: Beams Deep Review

結構技師「鋼結構設計」核心筆記



The Data

主考點題數佔
全科比例 (21/98)

全科出題排名 #2

近6年出題 5/6

21.4%

battery power

命題分析 (Battlefronts)



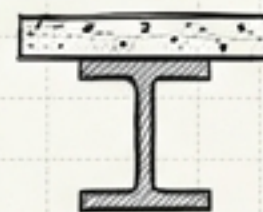
LTB 側扭挫屈 (12題)

$l_p/l_r/C_b$ 、彈性與非彈性 LTB — 絕對主戰場



塑性彎矩與斷面分類 (5題)

M_p/M_y / 形狀因子、寬厚比 λ_p/λ_r



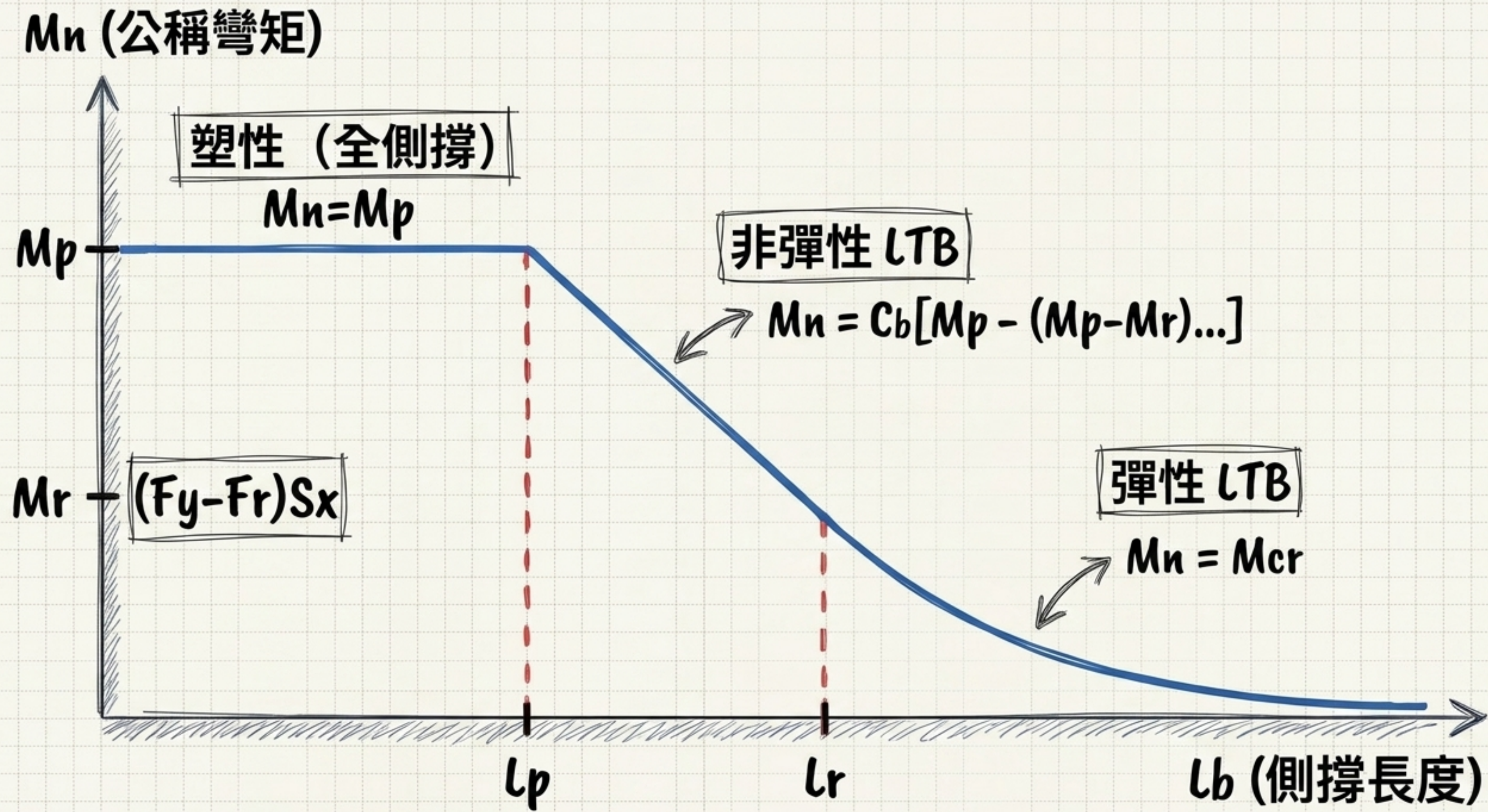
合成梁 / SRC (2題)

塑性應力分佈、PNA

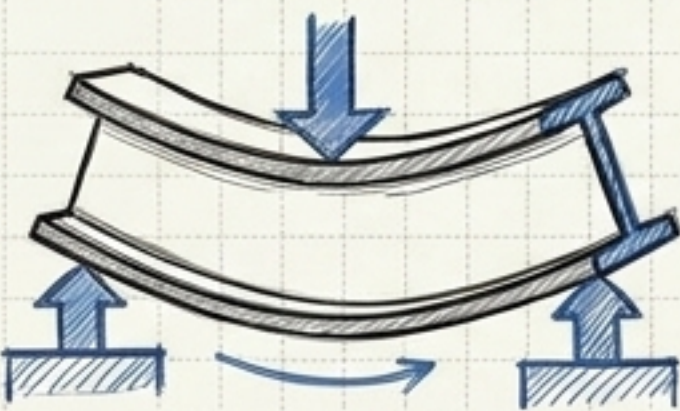
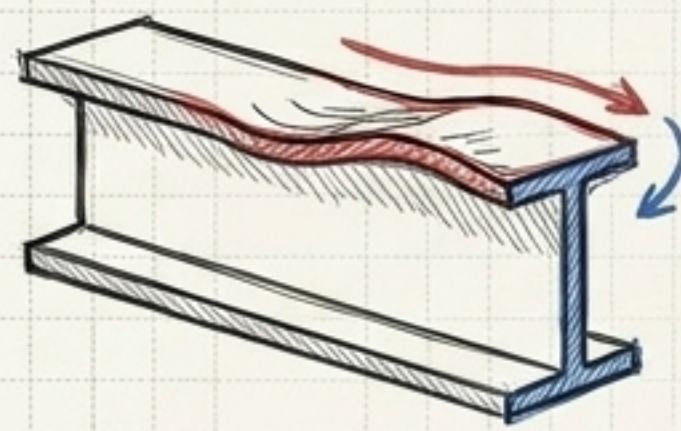
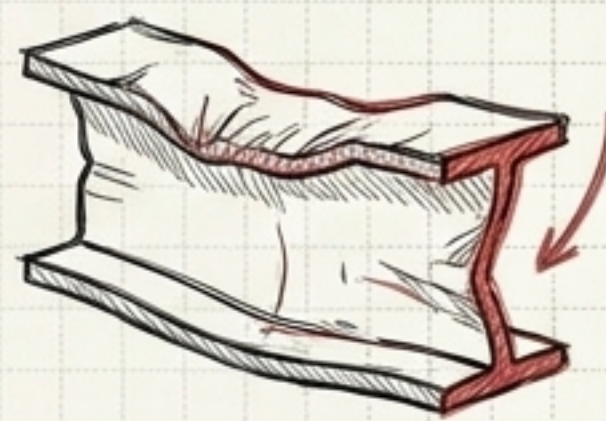


板梁與剪力 (2題)

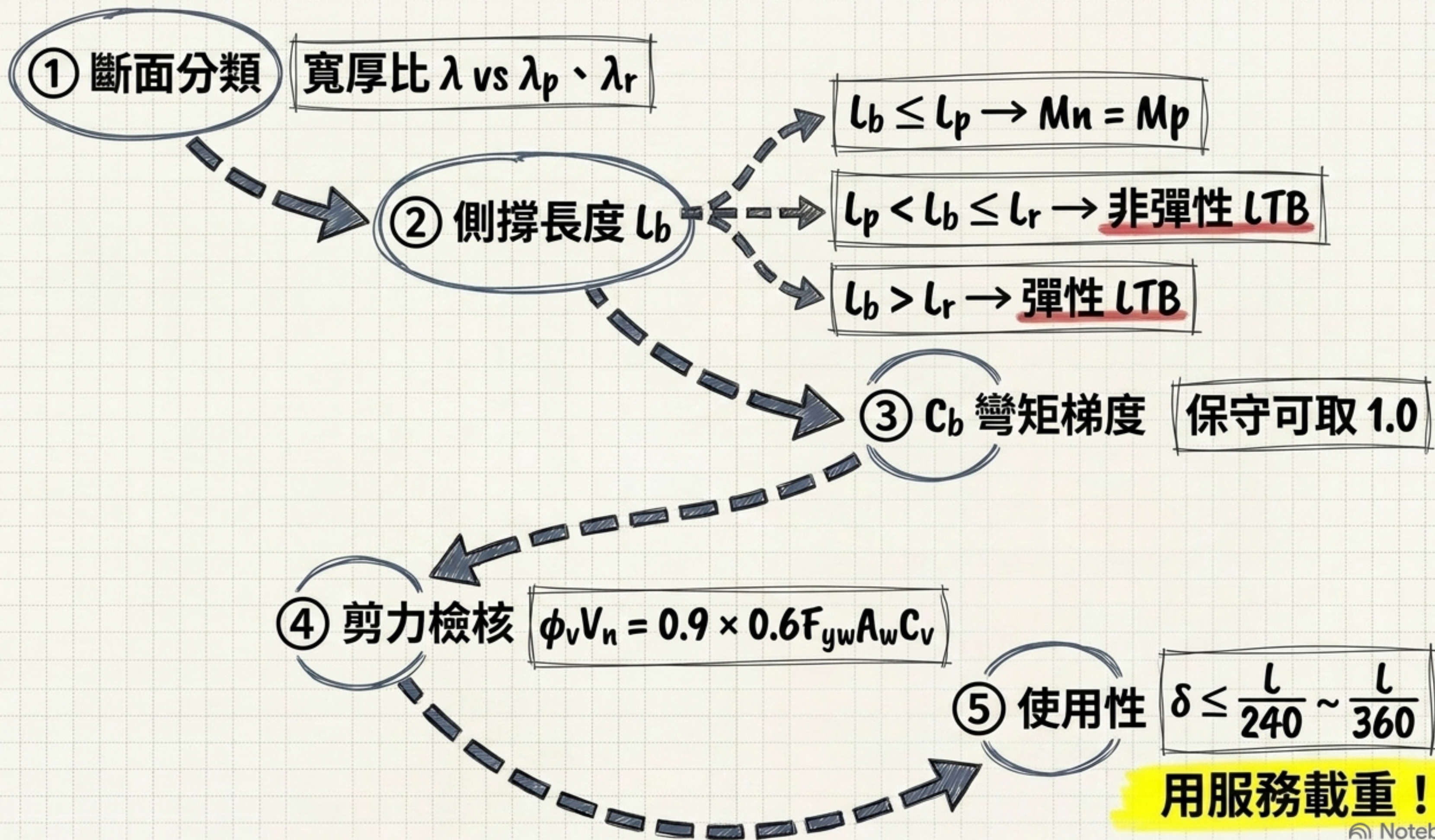
剪力挫屈、張力場、剪力流



斷面分類 (Cross-section Classification)

	結實 (Compact)	非結實 (Non-compact)	細長 (Slender)
Limit Checks	$\lambda \leq \lambda_p$	$\lambda_p < \lambda \leq \lambda_r$	$\lambda > \lambda_r$
Governing Behavior & Doodles	可達 M_p (耐震再加嚴 λ_{pd}) 	在 M_p 與 M_r 之間內插 	彈性局部挫屈控制 (Q 折減) 

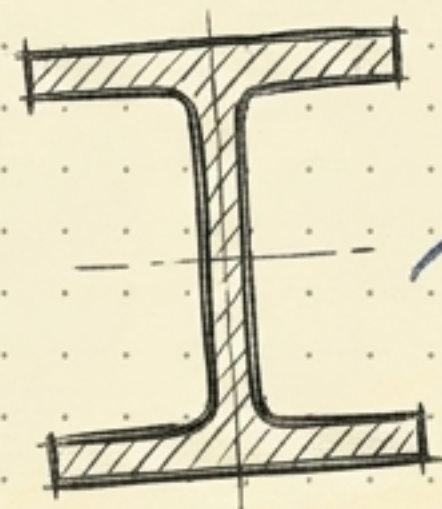
⚠ 翼板懸伸 $b = b_f/2$; 腹板取 h/t_w
 M_n 取「LTB、FLB、WLB」三者最小



$$M_p = F_y Z$$

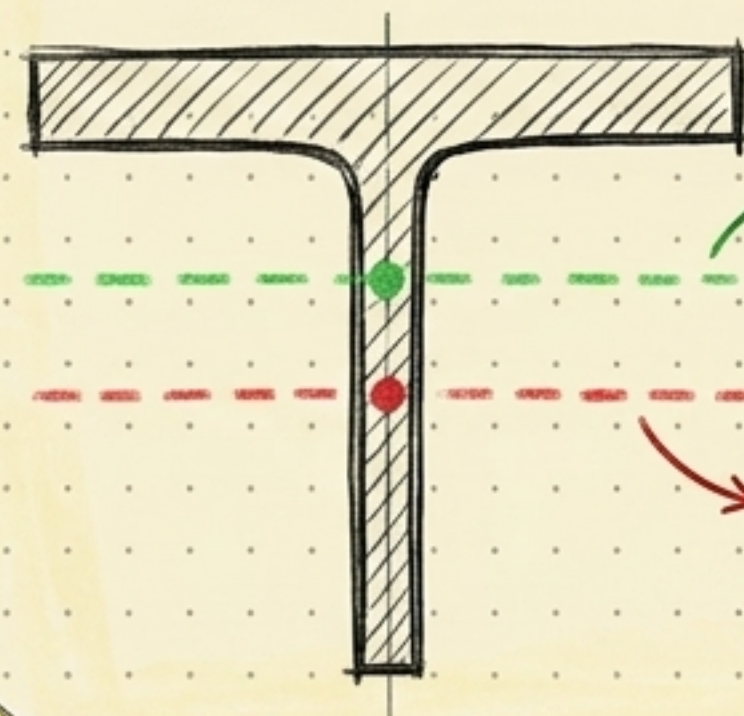
$$M_y = F_y S$$

$$\text{形狀因子 } f = \frac{Z}{S} = \frac{M_p}{M_y}$$



$$f \approx 1.10 - 1.18$$

非對稱斷面的陷阱 (Asymmetric Traps)



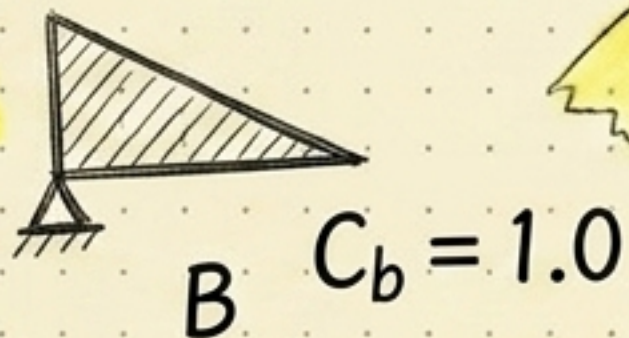
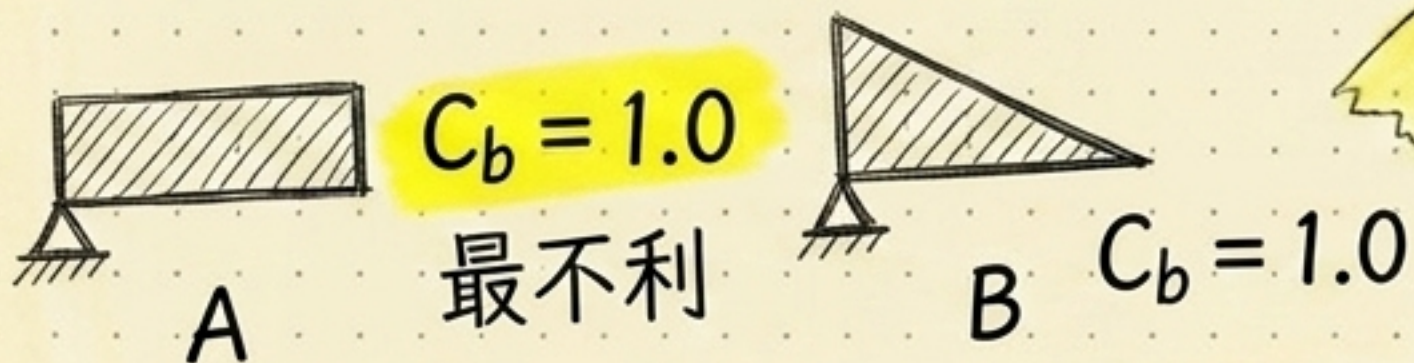
彈性中性軸 (ENA)

→ 過形心 (Centroid)

塑性中性軸 (PNA)

→ 等面積軸 (Equal Area)

$$C_b = \frac{12.5M_{\max}}{2.5M_{\max} + 3M_A + 4M_B + 3M_C}$$



M_1/M_2 雙曲率取正

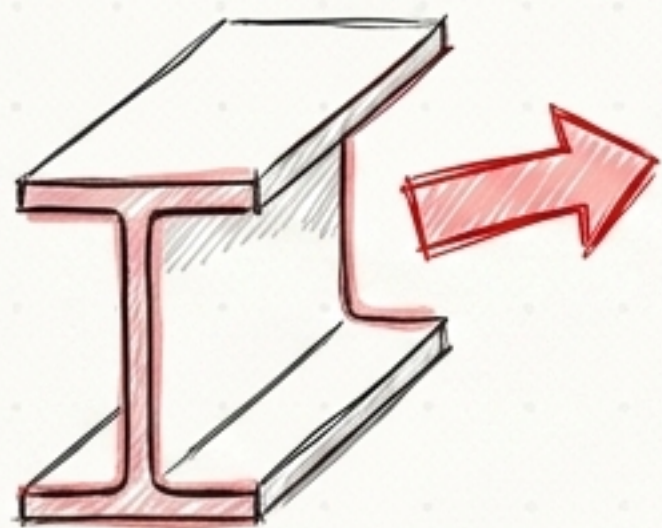
$$M_{cr} = C_b \frac{\pi}{L_b} \sqrt{\underbrace{EI_y GJ}_{\text{St. Venant 純扭}} + \underbrace{\left(\frac{\pi E}{L_b}\right)^2 I_y C_w}_{\text{翹曲扭轉 (Warping)}}}$$

$$L_p = 1.76 r_y \sqrt{\frac{E}{F_y}}$$

翹曲扭轉
(Warping)

✂ 剪力

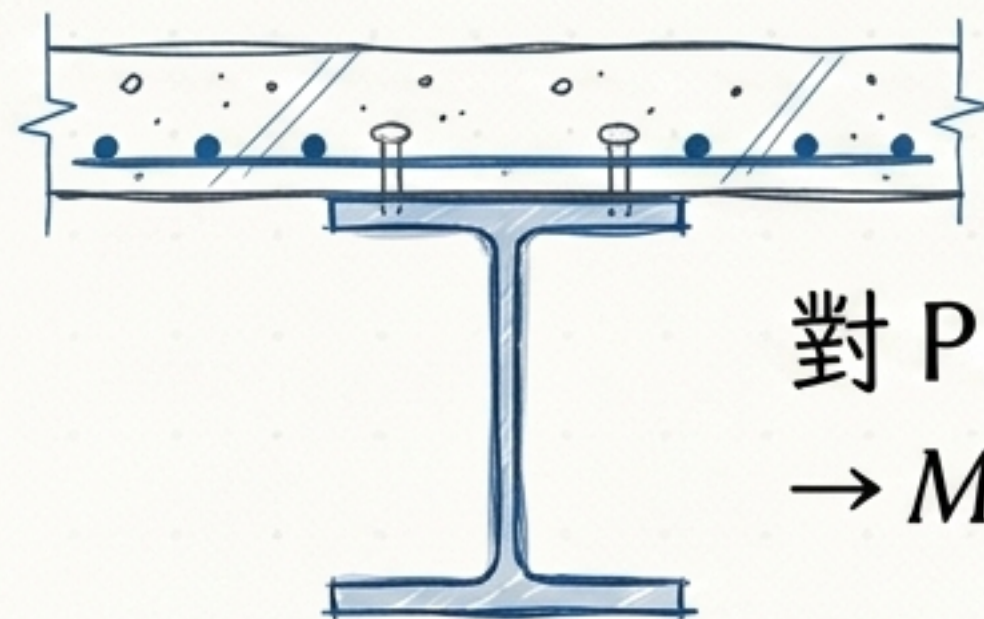
$$\phi_v V_n = 0.90 \times 0.6 F_{yw} A_w C_v$$



弱軸剪力由翼板
承擔: $A_w = 2 b_f t_f$

🧱 合成梁

$$C = \min(0.85 f'_c b_e t_s, F_y A_s, \Sigma Q_n)$$



對 PNA 取矩
 $\rightarrow M_n = C \cdot d_1 + \dots$

📏 撓度

$$\delta_{\max} = \frac{5wL^4}{384EI} \leq \frac{L}{240} \sim \frac{L}{360}$$

未乘載重係數！



高頻陷阱 Top 8

Panel 1 (The C_b Illusion)



Trap: 非彈性 LTB $\times C_b$
 $\rightarrow$ 巨大 M_n 值



Truth: 必須截斷於 M_p

$$M_n \leq M_p$$

Panel 2 (The F_r Ghost)



Trap: 用錯殘留應力 F_r



Truth:

銲接組合就是 1.16 !

軋軋 = 0.7

Panel 3 (C_b Overestimation)



純彎/懸臂計算複雜 C_b



純彎矩 / 懸臂自由端 = 1.0



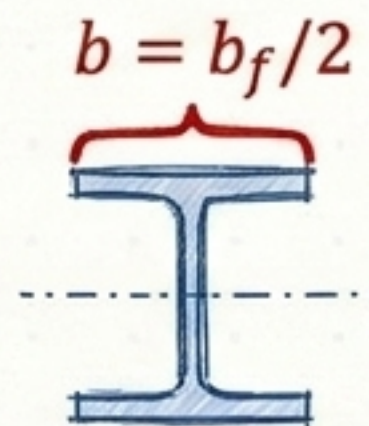
Panel 4 (Flange Width Error)



$$\lambda = \frac{b_f}{t_f}$$



懸伸肢 $b =$



分類錯 → 全題走錯公式

Panel 5 (Asymmetric PNA)



$$Z = \sum A \cdot (\text{距形心距離})$$



PNA = 等面積軸

M_y 用形心， M_p 用 PNA

Panel 6 (Amplified Deflection)



δ 使用 ~~$1.2D + 1.6L$~~



撓度屬使用性極限狀態

使用 $1.0D + 1.0L$

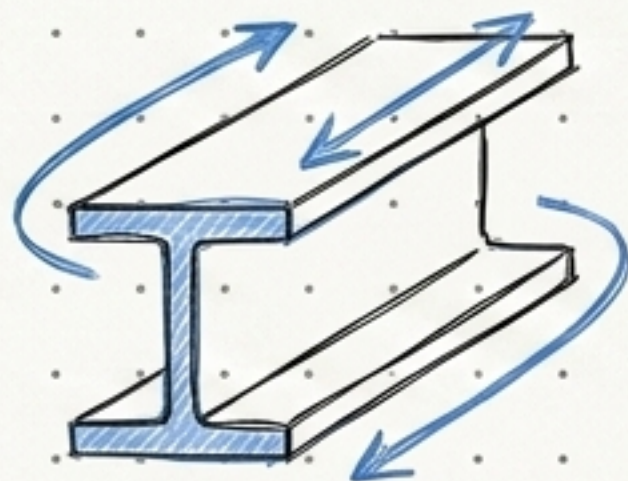
Panel 7 (Weak Axis Shear)



弱軸剪力用 $d \cdot t_w$



弱軸方向剪力流由『兩翼板』承擔



$$A_w = 2b_f \cdot t_f$$

Panel 8 (Composite PNA Guesswork)

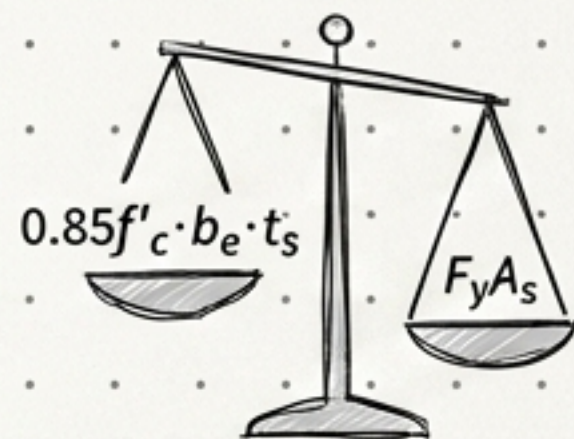


盲目猜測 PNA 在版內



未分況討論

先比較 $0.85f'_c \cdot b_e \cdot t_s$
與 $F_y A_s$ 決定 PNA 位置



21 歷史考題已掌握

- ☒ 12 LTB 側扭挫屈
- ☒ 5 塑性彎矩/斷面分類
- ☒ 2 合成梁/SRC
- ☒ 2 板梁與剪力

SS-2025-3,
SS-2015-1

LTB 公稱彎矩 互動計算器

M_p	M_r	L_p
L_r	L_b	C_b

使用 LTB 互動計算器驗證 M_p , M_r , L_p , L_r 曲線以完成複習作業